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Inventor(s)	Son; Won Sik et al.

Substrate treating apparatus and substrate treating method

Abstract

Provided is an apparatus for treating a substrate including: a processing vessel having a processing space; a support unit for supporting the substrate in the processing space and rotating the substrate; a liquid supply unit for supplying a processing liquid to the substrate; and a heating unit for heating the substrate, wherein the support unit includes: a spin chuck; a driver for rotating the spin chuck; a chuck pin installed on the spin chuck to be rotated together with the spin chuck; and a chuck pin moving unit for moving the chuck pin between a contact position wherein the chuck pin is in contact with a side portion of the substrate and an open position wherein the chuck pin is spaced apart from the side portion of the substrate.

Inventors: **Son; Won Sik (Gyeonggi-do, KR), Oh; Se Hoon (Chungcheongnam-do, KR), Kim; Jin Kyu (Gyeonggi-do, KR), Jung; In Ki (Gyeonggi-do, KR), Yu; Jeong Hyup (Gyeonggi-do, KR)**

Applicant: **SEMES CO., LTD.** (Chungcheongnam-do, KR)

Family ID: **1000008782482**

Assignee: **SEMES CO., LTD. (Chungcheongnam-Do, KR)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8172646	12/2011	Feng	451/367	H01L 21/68728
2017/0250097	12/2016	Hamada et al.	N/A	N/A
2019/0096737	12/2018	Abe	N/A	H01L 21/67017
2019/0198363	12/2018	Shimai et al.	N/A	N/A
2020/0251358	12/2019	Cha	N/A	H01L 21/67115
2021/0202280	12/2020	Kim et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2002-057205	12/2001	JP	N/A
2008-060579	12/2007	JP	N/A
2014-157936	12/2013	JP	N/A
2016-192545	12/2015	JP	N/A
2017-152686	12/2016	JP	N/A
2018-019103	12/2017	JP	N/A
2018-050014	12/2017	JP	N/A
2020-047719	12/2019	JP	N/A
2021-106237	12/2020	JP	N/A
2021-106257	12/2020	JP	N/A
10-1145775	12/2011	KR	N/A
10-1445775	12/2013	KR	N/A
10-2020-0122486	12/2019	KR	N/A
10-2208753	12/2020	KR	N/A
10-2248770	12/2020	KR	N/A
10-2021-0083092	12/2020	KR	N/A

OTHER PUBLICATIONS

Notice of Reasons for Refusal issued by the Japanese Patent Office on Nov. 21, 2023 in corresponding JP Patent Application No. 2022-163984. cited by applicant
Korean Office Action issued in corresponding KR Patent Application No. 10-2021-0136377, dated Feb. 21, 2025, with English translation. cited by applicant

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Primary Examiner: Hall, Jr.; Tyrone V

Attorney, Agent or Firm: Carter, DeLuca & Farrell LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2021-0136377 filed in the Korean Intellectual Property Office on Oct. 14, 2021, the entire contents of which are incorporated herein by reference.
TECHNICAL FIELD
(2) The present invention relates to a substrate treating apparatus and a substrate treating method.
BACKGROUND ART
(3) In order to manufacture a semiconductor device or a liquid crystal display, various processes, such as photography, ashing, ion implantation, thin film deposition, and cleaning, performed on a substrate. Among them, the etching process or the cleaning process is a process of removing unnecessary regions of a thin film formed on a substrate, or etching or cleaning foreign substances, particles, and the like, and requires high selectivity, high etch rate, and etch uniformity for the thin film, and as semiconductor devices are highly integrated, higher levels of etch selectivity and etch uniformity are required.
(4) In general, in the etching process or cleaning process of the substrate, a processing liquid treatment operation, a rinse treatment operation, and a drying treatment operation are sequentially performed. In one example, in the processing liquid treatment operation, a processing liquid for etching the thin film formed on the substrate or removing foreign substances on the substrate is supplied to the substrate to form a puddle, and then the puddle of the processing liquid is heated to promote the etching by the processing liquid, and in the rinse treatment operation, a rinse liquid, such as pure water, is supplied onto the substrate.
(5) The above-described processing liquid treatment operation is performed by placing the substrate on a support unit and supplying the processing liquid onto the substrate while rotating the support unit. The support unit is provided with chuck pins for supporting the side portion of the substrate to prevent the substrate from moving in the lateral direction of the support unit during rotation. The chuck pins move between a standby position that provides space for a substrate to be placed when the substrate is loaded or unloaded onto the support unit, and a support position that comes into contact with the side portion of the substrate as the process is performed while the substrate placed on the support unit is rotated. Accordingly, the space provided between the chuck pins placed in the standby position is wider than the space provided between the chuck pins placed in the support position.
(6) In general, when the puddle is formed on the substrate, a problem occurs in that the processing liquid flows down along the chuck pin due to the contact between the substrate and the chuck pin. In addition, there is a problem in that it is difficult to maintain a certain amount of liquid film due to the phenomenon in which the processing liquid flows down along the chuck pin.
SUMMARY OF THE INVENTION

(7) The present invention has been made in an effort to provide a substrate treating apparatus and a substrate treating method capable of efficiently treating a substrate.
(8) The present invention has also been made in an effort to provide a support unit in which a chuck pin freely moves during a process by rotating a substrate, and a substrate treating apparatus and method using the same.
(9) The object of the present invention is not limited thereto, and other objects not mentioned will be clearly understood by those of ordinary skill in the art from the following description.
(10) An exemplary embodiment of the present invention provides an apparatus for treating a substrate, the apparatus including: a processing vessel having a processing space; a support unit for supporting the substrate in the processing space and rotating the substrate; a liquid supply unit for supplying a processing liquid to the substrate supported by the support unit; and a heating unit for heating the substrate, in which the support unit includes: a spin chuck; a driver for rotating the spin chuck; a chuck pin installed on the spin chuck so as to be rotated together with the spin chuck; and a chuck pin moving unit for moving the chuck pin between a contact position in which the chuck pin is in contact with a side portion of the substrate and an open position in which the chuck pin is spaced apart from the side portion of the substrate, the chuck pin moving unit includes: a first driving module coupled to the chuck pin and rotated together with the spin chuck; and a second driving module which faces the first driving module and is not rotated together with the spin chuck, the first driving module includes a first magnetic body, and the second driving module includes a second magnetic body facing the first magnetic body, and a driving member for driving the second magnetic body in a vertical direction.
(11) The first driving module may move a position of the chuck pin according to a change in a position of the second magnetic body.
(12) Repulsive force may act between the first magnetic body and the second magnetic body.
(13) The first driving module may further include an arm member connecting the first magnetic body and the chuck pin, and the arm member may guide the movement of the chuck pin as the first magnetic body moves.
(14) The chuck pin may include a plurality of chuck pins, the second magnetic body may be provided in a ring shape, and the first driving module may be provided in a number corresponding to the plurality of chuck pins.
(15) The arm member may include: a first arm coupled to the chuck pin and extending in a direction perpendicular to a longitudinal direction of the chuck pin; a second arm coupled to the first arm; and a third arm for connecting the second arm and the first magnetic body, an elastic member may be coupled to one end of the second arm, and the elastic member may provide recovery force so that the chuck pin is moved from the open position to the contact position.
(16) The chuck pin moving unit may move the chuck pin while the spin chuck is rotating.
(17) The driver may rotate the spin chuck so that the substrate is rotated at a first speed, and the chuck pin moving unit may move the chuck pin so that the chuck pin is positioned at the open position while the substrate is rotating at the first speed.
(18) The driver may rotate the spin chuck so that the substrate is rotated at a second speed faster than the first speed, and the chuck pin moving unit may move the chuck pin so that the chuck pin is positioned at the contact position while the substrate is rotating at the second speed.
(19) The spin chuck may have a through-hole penetrating in a vertical direction, and the heating unit may heat a bottom surface of the substrate through the through-hole.
(20) The heating unit may include a laser.
(21) The spin chuck may include: a body portion; and an extension portion extending upwardly from an upper end of the body portion, and an area of the extension portion may gradually increase toward the top.
(22) Another exemplary embodiment of the present invention provides a method of treating a substrate, the method including: a first liquid supply operation of supplying a first liquid to a

substrate rotating at a first speed in an open state in which a chuck pin provided to support a side portion of the substrate is spaced apart from the side portion of the substrate and forming a first liquid film on the substrate; a liquid film heating operation of heating the first liquid film formed on the substrate in the open state, after the first liquid supply operation; and a second liquid supply operation of supplying a second liquid to the substrate rotating at a second speed in a contact state in which the chuck pin is in contact with the side portion of the substrate to support the side portion of the substrate, after the liquid film heating operation.

(23) The first liquid and the second liquid may be the same.

(24) The first liquid may be an aqueous solution of phosphoric acid.

(25) The change from the open state to the contact state may be made while the substrate is rotating.

(26) The second speed may be faster than the first speed.

(27) The amount of second liquid supplied per unit time in the second liquid supply operation may be greater than the amount of first liquid supplied per unit time in the first liquid supply operation.

(28) In the liquid film heating operation, the substrate may be rotated at the first speed, and the first liquid may not be supplied onto the substrate.

(29) Still another exemplary embodiment of the present invention provides a method of treating a substrate by using the apparatus of treating the substrate, the method including: a first liquid supply operation of supplying a first liquid to the substrate rotating at a first speed in an open state in which the chuck pin is positioned at the open position to form a first liquid film on the substrate; a liquid film heating operation of heating the first liquid film formed on the substrate in the open state of the chuck pin, after the first liquid supply operation; and a second liquid supply operation of supplying a second liquid to the substrate rotating at a second speed faster than the first speed in a contact state in which the chuck pin is positioned at the contact position, after the liquid film heating operation, in which the change of the chuck pin from the open state to the contact state is performed while the substrate is rotating.

(30) According to the exemplary embodiment of the present invention, it is possible to efficiently treat a substrate.

(31) Further, according to the exemplary embodiment of the present invention, the chuck pin may be freely moved while the process is performed by rotating the substrate.

(32) Further, according to the exemplary embodiment of the present invention, it is possible to prevent the processing liquid from flowing down to the chuck pin.

(33) Further, according to the exemplary embodiment of the present invention, the processing liquid puddle formed on the substrate may be maintained as a liquid film of a predetermined amount or more.

(34) Further, according to the exemplary embodiment of the present invention, the chuck pin may be constantly moved regardless of whether the support unit supporting the substrate is rotated or is not rotated.

(35) Further, according to the exemplary embodiment of the present invention, it is possible to adjust the movement stroke of the chuck pin during the process.

(36) Further, according to the exemplary embodiment of the present invention, through the support unit in which a hollow space is formed, it is possible to apply various types of heat sources and cooling systems of various heat sources.

(37) The effect of the present invention is not limited to the foregoing effects, and those skilled in the art may clearly understand non-mentioned effects from the present specification and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top plan view illustrating a substrate treating facility according to an exemplary embodiment of the present invention.

(2) FIG. **2** is a cross-sectional view illustrating a substrate treating apparatus provided in a process chamber of FIG. **1**.

(3) FIG. **3** is a cross-sectional view illustrating a chuck pin and a chuck pin moving unit according to an exemplary embodiment of the present invention.

(4) FIG. **4** is a perspective view illustrating a second magnetic body of the chuck pin moving unit of FIG. **3**.

(5) FIG. **5** is a top plan view illustrating a state in which the chuck pin is moved during rotation of a spin chuck according to the exemplary embodiment of the present invention.

(6) FIG. **6** is an operational view illustrating a process in which the chuck pin of FIG. **5** is moved to an open position.

(7) FIG. **7** is an operational view illustrating a process in which the chuck pin of FIG. **5** is moved to a contact position.

(8) FIG. **8** is a diagram schematically illustrating a relationship between a rotating first magnetic body and a non-rotating second magnetic body of FIG. **4**.

(9) FIG. **9** is a flowchart of a substrate treating method according to an exemplary embodiment of the present invention.

(10) FIGS. **10** to **13** are diagrams sequentially illustrating the substrate treating method of FIG. **9**.

DETAILED DESCRIPTION

(11) Hereinafter, an exemplary embodiment of the present invention will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the invention are illustrated. However, the present invention can be variously implemented and is not limited to the following exemplary embodiments. In addition, in describing an exemplary embodiment of the present invention in detail, if it is determined that a detailed description of a related well-known function or configuration may unnecessarily obscure the gist of the present invention, the detailed description thereof will be omitted. In addition, the same reference numerals are used throughout the drawings for parts having similar functions and actions.

(12) In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. It will be appreciated that terms “including” and “having” are intended to designate the existence of characteristics, numbers, operations, operations, constituent elements, and components described in the specification or a combination thereof, and do not exclude a possibility of the existence or addition of one or more other characteristics, numbers, operations, operations, constituent elements, and components, or a combination thereof in advance.

(13) Singular expressions used herein include plurals expressions unless they have definitely opposite meanings in the context. Accordingly, shapes, sizes, and the like of the elements in the drawing may be exaggerated for clearer description.

(14) An expression, “and/or” includes each of the mentioned items and all of the combinations including one or more of the items. Further, in the present specification, “connected” means not only when member A and member B are directly connected, but also when member A and member B are indirectly connected by interposing member C between member A and member B.

(15) The exemplary embodiment of the present invention may be modified in various forms, and the scope of the present invention should not be construed as being limited to the following exemplary embodiments. The present exemplary embodiment is provided to more completely explain the present invention to those skilled in the art. Therefore, the shapes of elements in the drawings are exaggerated to emphasize clearer descriptions.

(16) In the present exemplary embodiment, a process of etching a substrate by using a processing liquid will be described as an example. However, the present exemplary embodiment is not limited to the etching process, and is variously applicable to substrate treating processes using liquids, such as a cleaning process, an ashing process, and a developing process.
(17) Here, the substrate is a comprehensive concept including all substrates used for manufacturing semiconductor devices, Flat Panel Displays (FPDs), and other articles in which circuit patterns are formed on thin films. Examples of the substrate W include a silicon wafer, a glass substrate, and an organic substrate.
(18) Hereinafter, an example of the present invention will be described in detail with reference to FIGS. **1** to **14**.
(19) FIG. **1** is a top plan view illustrating a substrate treating facility **1** according to an exemplary embodiment of the present invention. Referring to FIG. **1**, the substrate treating facility **1** includes an index module **10** and a process processing module **20**. The index module **10** includes a load port **120** and a transfer frame **140**. The load port **120**, the transfer frame **140**, and the process processing module **20** may be sequentially arranged in series.
(20) Hereinafter, a direction in which the load port **120**, the transfer frame **140**, and the process processing module **20** are arranged is called to as a first direction **12**, and a direction perpendicular to the first direction **12** when viewed from the top is called a second direction **14**, and a direction perpendicular to a plane including the first direction **12** and the second direction **14** is called a third direction **16**.
(21) A carrier **18** in which a substrate W is accommodated is seated on the load port **120**. The load port **120** is provided in plurality, and the plurality of load ports **120** is arranged in series in the second direction **14**. The number of load ports **120** may be increased or decreased according to process efficiency of the process processing module **20** and a condition of foot print, and the like. A plurality of slots (not illustrated) for accommodating the plurality of substrates W in a state where the substrates W are arranged horizontally with respect to the ground may be formed in the carrier **18**. As the carrier **18**, a Front Opening Unified Pod (FOUP) may be used.
(22) The process module **20** includes a buffer unit **220**, a transfer chamber **240**, and a process chamber **260**.
(23) The transfer chamber **240** is disposed so that a longitudinal direction thereof is parallel to the first direction **12**. A plurality of process chambers **260** may be disposed on one or both sides of the transfer chamber **240**. At one side and the other side of the transfer chamber **240**, the plurality of process chambers **260** may be provided to be symmetrical with respect to the transfer chamber **240**. Some of the plurality of process chambers **260** are disposed along the longitudinal direction of the transfer chamber **240**. Further, some of the process chambers **260** are disposed to be stacked with each other. That is, the plurality of process chambers **260** may be disposed in an arrangement of A×B at one side of the transfer chamber **240**. Here, A is the number of process chambers **260** provided in a line along the first direction **12**, and B is the number of process chambers **260** provided in a line along the third direction **16**. When four or six process chambers **260** are provided at one side of the transfer chamber **240**, the process chambers **260** may be disposed in an arrangement of 2×2 or 3×2. The number of process chambers **260** may be increased or decreased. Unlike the foregoing, the process chamber **260** may be provided only to one side of the transfer chamber **240**. In addition, the process chamber **260** may be provided as a single layer on one side and both sides of the transfer chamber **240**.
(24) The buffer unit **220** is disposed between the transfer frame **140** and the transfer chamber **240**. The buffer unit **220** provides a space in which the substrate W stays before the substrate W is transferred between the transfer chamber **240** and the transfer frame **140**. A slot (not illustrated) in which the substrate W is placed is provided inside the buffer unit **220**. A plurality of slots (not illustrated) is provided so as to be spaced apart from each other in the third direction **16**. The buffer unit **220** has an open side facing the transfer frame **140**. The buffer unit **220** has an open side facing

the transfer chamber **240**.

(25) The transfer frame **1400** transfers the substrate W between the carrier **18** seated on the load port **1200** and the buffer unit **220**. In the transfer frame **140**, an index rail **142** and an index robot **144** are provided. A longitudinal direction of the index rail **142** is provided to be parallel to the second direction **14**. The index robot **144** is installed on the index rail **142**, and linearly moves in the second direction **14** along the index rail **142**. The index robot **144** includes a base **144***a*, a body **144***b*, and an index arm **144***c*. The base **144***a* is installed to be movable along the index rail **142**. The body **144***b* is coupled to the base **144***a*. The body **144***b* is provided to be movable in the third direction **16** on the base **144***a*. Further, the body **144***b* is provided to be rotatable on the base **144***a*. The index arm **144***c* is coupled to the body **144***b* and is provided to be movable forwardly and backwardly with respect to the body **144***b*. A plurality of index arms **144***c* is provided to be individually driven. The index arms **144***c* are disposed to be stacked in the state of being spaced apart from each other in the third direction **16**. A part of the index arms **144***c* may be used when the substrate W is transferred from the process processing module **20** to the carrier **18**, and another part of the plurality of index arms **144***c* may be used when the substrate W is transferred from the carrier **18** to the process processing module **20**. This may prevent particles generated from the substrate W before the process processing from being attached to the substrate W after the process processing in the process of loading and unloading the substrate W by the index robot **144**.

(26) The transfer chamber **2400** transfers the substrate W between the buffer unit **2200** and the process chamber **260**, and between the process chambers **260**. A guide rail **242** and a main robot **244** are provided to the transfer chamber **240**. The guide rail **242** is disposed so that its longitudinal direction is parallel to the first direction **12**. The main robot **244** is installed on the guide rail **242** and linearly moved along the first direction **12** on the guide rail **242**. The main robot **244** includes a base **244***a*, a body **244***b*, and a main arm **244***c*. The base **244***a* is installed to be movable along the guide rail **242**. The body **244***b* is coupled to the base **244***a*. The body **244***b* is provided to be movable in the third direction **16** on the base **244***a*. Further, the body **244***b* is provided to be rotatable on the base **244***a*. The main arm **244***c* is coupled to the body **244***b*, and provided to be movable forwardly and backwardly with respect to the body **244***b*. A plurality of main arms **244***c* is provided to be individually driven. The main arms **244** are disposed to be stacked in the state of being spaced apart from each other in the third direction **16**.

(27) A substrate treating apparatus **300** performing a liquid processing process on the substrate W is provided to the process chamber **260**. The substrate treating apparatus **300** may have a different structure depending on the type of liquid processing process to be performed. Contrary to this, the substrate treating apparatus **300** within each process chamber **260** may have the same structure. Optionally, the plurality of process chambers **260** is divided into a plurality of groups, and the substrate treating apparatuses **300** within the process chambers **260** belong to the same group may have the same structure, and the substrate treating apparatuses **300** within the process chambers **260** belong to the different groups may have the different structures.

(28) FIG. **2** is a cross-sectional view illustrating a substrate treating apparatus provided in a process chamber of FIG. **1**, FIG. **3** is a cross-sectional view illustrating a chuck pin and a chuck pin moving unit according to an exemplary embodiment of the present invention, FIG. **4** is a perspective view illustrating a second magnetic body of the chuck pin moving unit of FIG. **3**, FIG. **5** is a top plan view illustrating a state in which the chuck pin is moved during rotation of a spin chuck according to the exemplary embodiment of the present invention, FIG. **6** is an operational view illustrating a process in which the chuck pin of FIG. **5** is moved to an open position, FIG. **7** is an operational view illustrating a process in which the chuck pin of FIG. **5** is moved to a contact position, and FIG. **8** is a diagram schematically illustrating a relationship between a rotating first magnetic body and a non-rotating second magnetic body of FIG. **4**.

(29) Referring to FIG. **2**, the substrate treating apparatus **300** includes a processing vessel **320**, a substrate support unit **340**, a lifting unit **360**, a liquid supply unit **390**, and a heating unit **500**.

(30) The substrate treating apparatus **300** may include a chamber **310**. The chamber **310** provides a sealed inner space. The processing vessel **320** is disposed in the inner space of the chamber **310**. A fan filter unit **315** is installed at the upper portion of the chamber **310**. The fan filter unit **315** generates a vertical airflow in the chamber **310**. The fan filter unit **315** generates a downward airflow in the chamber **310**. The fan filter unit **315** is a device in which a filter and an air supply fan are modularized as one unit, and which filters high-humidity outdoor air and supplies the filtered air to the inside of the chamber **310**. The high-humidity outdoor air passes through the fan filter unit **315** and is supplied into the chamber **310**, and forms a vertical airflow in the inner space of the chamber **310**. This vertical airflow provides a uniform airflow in the upper portion of the substrate W. Contaminants (for example, fumes) generated in the process of treating the substrate W are discharged to the outside of the processing vessel **320** together with the air included in the vertical airflow, and through this, the inside of the processing vessel **320** may maintain a high degree of cleanliness.

(31) The treating vessel **320** has a cylindrical shape with an open top. The processing vessel **320** includes a first recovery container **321** and a second recovery container **322**. The recovery containers **321** and **322** recovers different processing liquids among the processing liquids used for the process. The first recovery container **321** is provided in an annular ring shape surrounding the substrate support unit **340**. The second recovery container **322** is provided in the shape of an annular ring surrounding the substrate support unit **340**. In one exemplary embodiment, the first recovery container **321** is provided in an annular ring shape surrounding the second recovery container **322**. The second recovery container **322** may be provided by being inserted into the first recovery container **321**. The height of the second recovery container **322** may be higher than the height of the first recovery container **321**. The second recovery container **322** may include a first guard part **326** and a second guard part **324**. The first guard part **326** may be provided at the top of the second recovery container **322**. The first guard part **326** is formed to extend toward the substrate support unit **340**, and the first guard part **326** may be formed to be inclined upward toward the substrate support unit **340**. In the second recovery container **322**, the second guard part **324** may be provided at a position spaced downward from the first guard part **326**. The second guard part **324** is formed to extend toward the substrate support unit **340**, and the second guard part **324** may be formed to be inclined upward toward the substrate support unit **340**. The space between the first guard part **326** and the second guard part **324** functions as a first inlet **324***a* through which the processing liquid flows. A second inlet **322***a* is provided at a lower portion of the second guard part **324**. The first inlet **324***a* and the second inlet **322***a* may be located at different heights. A hole (not illustrated) is formed in the second guard part **324** so that the processing liquid flowing into the first inlet **324***a* flows into a second recovery line **322***b* provided in the lower portion of the second recovery container **322**. A hole (not illustrated) of the second guard part **324** may be formed at a position with the lowest height in the second guard part **324**. The processing liquid recovered to the first recovery container **321** is configured to flow into the first recovery line **321***b* connected to the bottom surface of the first recovery container **321**. The processing liquids introduced into the recovery containers **321** and **322** may be provided to an external processing liquid recycling system (not illustrated) through the recovery lines **321***b* and **322***b*, respectively, to be re-used.

(32) The lifting unit **360** linearly moves the processing vessel **320** up and down. As an example, the lifting unit **360** is coupled to the second recovery container **322** of the processing vessel **320** and moves the second recovery container **322** up and down, so that the relative height of the processing vessel **320** with respect to the substrate support unit **340** may be changed. The lifting unit **360** includes a bracket **362**, a moving shaft **364**, and a driver **366**. The bracket **362** is fixedly installed on the outer wall of the processing vessel **320**, and a moving shaft **364**, which is moved in the vertical direction by the driver **366**, is fixedly coupled to the bracket **362**. The second recovery container **322** of the processing vessel **320** is lowered so that the upper portion of the substrate

support unit **340** protrudes toward the upper portion of the processing vessel **320**, specifically, protrudes higher than the first guard part **326** when the substrate W is loaded into the substrate support unit **340** or unloaded from the substrate support unit **340**. Further, when the process proceeds, the height of the treating vessel **320** is adjusted so that the processing liquid is introduced into the predetermined recovery container **321** and **322** depending on the type of the processing liquid supplied to the substrate W. Optionally, the lifting unit **360** may also move the substrate support unit **340** in the vertical direction instead of the processing vessel **320**. Optionally, the lifting unit **360** may move the entire processing vessel **320** to be movable up and down in the vertical direction. The lifting unit **360** is provided to adjust the relative heights of the processing vessel **320** and the substrate support unit **340**, and if it is a configuration capable of adjusting the relative heights of the processing vessel **320** and the substrate support unit **340**, exemplary embodiments of the processing vessel **320** and the lifting unit **360** may be provided in various structures and methods depending on the design.
(33) The liquid supply unit **390** is configured to discharge a chemical liquid from an upper portion of the substrate W to the substrate W, and may include one or more chemical liquid discharge nozzles. The liquid supply unit **390** may pump the chemical liquid stored in a storage tank (not illustrated) to discharge the chemical liquid to the substrate W through the chemical liquid discharge nozzle. The liquid supply unit **390** may include a driving unit (not illustrated) to be movable between a process position facing the central region of the substrate W and a standby position outside the substrate W.
(34) The chemical liquid supplied from the liquid supply unit **390** to the substrate W may be various depending on the substrate treatment process. When the substrate treatment process is a silicon nitride film etching process, the chemical liquid may be a chemical liquid including phosphoric acid (H.sub.3PO.sub.4). The liquid supply unit **390** may further include a deionized water (DIW) supply nozzle for rinsing the surface of the substrate after the etching process, and an isopropyl alcohol (IPA) discharge nozzle and a nitrogen (N.sub.2) discharge nozzle for performing a drying process after rinsing. Although not illustrated, the liquid supply unit **390** may include a nozzle moving member (not illustrated) which is capable of supporting the chemical liquid discharge nozzle and moving the chemical liquid discharge nozzle. The nozzle moving member (not illustrated) may include a support shaft (not illustrated), an arm (not illustrated), and a driver (not illustrated). The support shaft (not illustrated) is located at one side of the treating vessel **320**. The support shaft (not illustrated) includes a rod shape of which a longitudinal direction faces the third direction. The support shaft (not illustrated) is provided to be rotatable by the driver (not illustrated). The arm (not illustrated) is coupled to an upper end of the support shaft (not illustrated). The arm (not illustrated) may be extended vertically from the support shaft (not illustrated). The chemical liquid discharge nozzle is fixedly coupled to the distal end of the arm (not illustrated). According to the rotation of the support shaft (not illustrated), the chemical liquid discharge nozzle is capable of swing together with the arm (not illustrated). The chemical liquid discharge nozzle may be swing-moved to move to the process position and the standby position. Optionally, the support shaft (not illustrated) may be provided to be movable up and down. Further, the arm (not illustrated) may be provided to be movable forward and backward toward the longitudinal direction thereof.
(35) The substrate support unit **340** supports the substrate W and rotates the substrate W during the process progress. The substrate support unit **340** includes a spin chuck **342**, a window member **348**, a driving member **349**, a chuck pin **346**, and a chuck pin moving unit **400**.
(36) The chuck pin **346** is coupled to the spin chuck **342**. The substrate W is spaced apart from the upper surface of the spin chuck **342** by the chuck pin **346** coupled to the spin chuck **342**. The substrate W is rotated together with the spin chuck **342** while being supported by the chuck pin **346** coupled to the spin chuck **342**.
(37) The spin chuck **342** is provided in the shape of a tank having an open top and an open bottom.

The spin chuck **342** includes a through-hole **342***a* penetrating through the upper surface and the lower surface. The spin chuck **342** includes the through-hole **342***a* penetrating in the vertical direction. In this case, the vertical direction may refer to an axial direction of the spin chuck **342** or a direction parallel to a rotational axis direction of the spin chuck **342**. The heating unit **500** to be described later is disposed in the through-hole **342***a*.
(38) The spin chuck **342** includes a body portion **3421** and an extension portion **3422** extending upwardly from the body portion **3421**. The body portion **3421** and the extension portion **3422** are integrally formed. The through-hole **342***a* is formed to pass through both the body portion **3421** and the extension portion **3422**. The body portion **3421** is formed to have the same area. The body portion **3421** is formed to have the same inner diameter. The through-hole **342***a* in the body portion **3421** is formed to have the same diameter. The extension portion **3422** is formed to gradually increase in area from the body portion **3421** in the upper direction. The inner diameter of the extension portion **3422** is formed to increase in the upper direction. The through-hole **342***a* in the extension portion **3422** is formed to increase in diameter in the upper direction. The heating unit **500** to be described later is disposed inside the body **3421**, and a laser beam generated from the heating unit **500** is emitted to the substrate W through the extension portion **3422**. The extension **3422** may be formed in a size that does not interfere with the laser beam. Through this, the laser beam generated by the heating unit **500** may be emitted to the substrate W without being interfered by the spin chuck **342**. The spin chuck **342** may be disposed under the window member **348** to be described later. The spin chuck **342** supports the edge region of the window member **348**. A connection portion between the spin chuck **342** and the window member **348** may have a sealing structure so that the chemical liquid supplied to the substrate W does not penetrate into the heating unit **500**.
(39) The window member **348** is located under the substrate W. The window member **348** is provided under the substrate W supported on the chuck pin **346**. The window member **348** may be provided in a shape substantially corresponding to the substrate W. For example, when the substrate W is a circular wafer, the window member **348** may be provided in a substantially circular shape. The window member **348** may have a larger diameter than the substrate. However, the present invention is not limited thereto, and the window member **348** may have the same diameter as that of the substrate W or may be formed to have a smaller diameter than that of the substrate W. A hole **3481** in which the chuck pin **346** is disposed is formed in the window member **348**. The chuck pin **346** may pass through the hole **3481** of the window member **348**. A diameter of the hole **3481** of the window member **348** may be larger than a diameter of the chuck pin **346**. Through this, the chuck pin **346** may move in the inside of the hole **3481** of the window member **348**. At this time, the chuck pin **346** is moved in a direction perpendicular to the rotation axis direction of the spin chuck **342**.
(40) The window member **348** may be made of a material having high light transmittance. Accordingly, the laser beam emitted from the beam emitting unit **400** may pass through the window member **348**. The window member **348** may be made of a material having excellent corrosion resistance so as not to react with the chemical liquid. For this purpose, the material of the window member **348** may be, for example, quartz, glass, or sapphire. The window member **348** is a configuration that allows the laser beam to pass through and reaches the substrate W, and protects the configuration of the substrate support member **340** from a chemical liquid, and may be provided in various sizes and shapes according to design.
(41) A support pin **347** may be coupled to the window member **348**. A plurality of support pins **347** may be provided. The support pin **347** may be provided in an edge region of the window member **348**. The plurality of support pins **347** may be disposed to be spaced apart from each other along the edge region of the window member **348**. The support pin **347** may be provided to protrude upwardly from the upper surface of the window member **348**. The support pin **347** may support the lower surface of the substrate W to separate the substrate W from the window member **348**.

(42) The driving member **349** is coupled to the spin chuck **342**. The driving member **349** rotates the spin chuck **342**. Any driving member **349** may be used as long as the driving member **349** is capable of rotating the spin chuck **342**. For example, the driving member **349** may be provided in a hollow motor. According to the exemplary embodiment, the driving member **349** may include a stator (not illustrated) and a rotor (not illustrated). The stator may be provided fixed at one position, and the rotor may be coupled to the spin chuck **342**. The rotor may be coupled to the bottom of the spin chuck **349** to rotate the spin chuck **342**. When a hollow motor is used as the driving member **349**, the narrower the bottom of the spin chuck **349** is provided, the smaller the hollow of the hollow motor can be selected, thereby reducing the manufacturing cost. According to the exemplary embodiment, a cover member (not illustrated) for protecting the driving member **349** from the chemical liquid may be further included.
(43) The chuck pin **346** is installed on the spin chuck **342**. The chuck pin **346** is installed on the extension portion **3422** of the spin chuck **342**. The chuck pin **346** rotates with the spin chuck **342**. A plurality of chuck pins **346** is provided. The plurality of chuck pins **346** is spaced apart from each other. The plurality of chuck pins **346** may be arranged in a circular shape when combined. The plurality of chuck pins **346** is provided along the edge of the through-hole **342***a* formed in the extension portion **3422**. The chuck pin **346** supports the side portion of the substrate W. The chuck pin **346** grips the substrate W. The chuck pin **3446** separates the substrate W from the window member **348** by a predetermined distance. At least a portion of the chuck pin **346** is received in the hole **3481** of the window member **348**. The chuck pin **346** is provided movably inside the hole **3481** of the window member **348**. The chuck pin **346** is coupled to the chuck pin moving unit **400** to be described later. The chuck pin **346** is provided to be movable by the chuck pin moving unit **400**. The chuck pin **342** is provided to be movable between a contact position in which the chuck pin **342** is in contact with the side portion of the substrate W and an open position in which the chuck pin **342** is spaced apart from the side portion of the substrate W.
(44) Referring to FIG. **3**, the chuck pin moving unit **400** is coupled to one end of the chuck pin **346** to move the chuck pin **346**. The chuck pin moving unit **400** moves the chuck pin **346** in a direction perpendicular to the rotation axis direction of the spin chuck **342**. The chuck pin moving unit **400** moves the chuck pin **346** to move between the contact position in which the chuck pin **342** is in contact with the side portion of the substrate W and the open position in which the chuck pin **342** is spaced apart from the side portion of the substrate W. The chuck pin moving unit **400** moves the chuck pin **342** while the spin chuck **342** is rotating. The chuck pin moving unit **400** moves the chuck pin **342** while the substrate W rotates.
(45) The chuck pin moving unit **400** includes a first driving module **420** and a second driving module **440**. The first driving module **420** and the second driving module **440** are combined to move the chuck pin **346**.
(46) The first driving module **420** is coupled to the chuck pin **346**. The first driving module **420** is accommodated in the spin chuck **432**. The first driving module **420** is accommodated in the extension portion **3422** of the spin chuck **342**. An accommodating space in which the first driving module **420** is accommodated is formed inside the extension portion **3422**. The accommodation space inside the extension **3422** may be provided as a groove. The first driving module **420** is rotated together with the spin chuck **432**.
(47) The first driving module **420** includes a first magnetic body **422** and an arm member **424**. The first magnetic body **420** is provided as a permanent magnet. The first magnetic body **422** faces a second magnetic body **442** of the second driving module **440** to be described later. The first magnetic body **422** is spaced apart from the second magnetic body **442**. The first magnetic body **422** overlaps at least a portion of the second magnetic body **422** in the direction of the rotation axis of the spin chuck **342**. The first magnetic body **422** and the second magnetic body **442** are provided so that the same polarities are opposite to each other. The polarity provided on the lower portion of the first magnetic body **422** has the same polarity as the polarity provided on the upper portion of

the first magnetic body **442**. For example, when the N polarity is disposed on the lower portion of the first magnetic body **422**, the N polarity is disposed on the upper portion of the second magnetic body **442**, and when the S polarity is disposed on the lower portion of the first magnetic body **422**, the S polarity is disposed on the upper portion of the second magnetic body **442**. Repulsive force acts between the first magnetic body **422** and the second magnetic body **442**. The first magnetic body **422** is moved in the direction of the rotation axis of the spin chuck **342** by the repulsive force with the second magnetic body **442**.

(48) The arm member **424** connects the chuck pin **346** and the first magnetic body **422**. One end of the arm member **424** is coupled to the chuck pin **346**, and the other end of the arm member **424** is coupled to the first magnetic body **422**. The arm member **424** guides the movement of the chuck pin **342** as the first magnetic body **422** moves by the repulsive force with the second magnetic body **422**.

(49) The arm member **424** may include a first arm **4242**, a second arm **4244**, and a third arm **4246**. The first arm **4242** is coupled to the chuck pin **346**. One end of the first arm **4242** is coupled to the chuck pin **345**, and the other end of the first arm **4242** is coupled to the second arm **4244**. The first arm **4242** extends in a direction perpendicular to the longitudinal direction of the chuck pin **346**. The first arm **4242** is disposed inside the chuck pin **346**.

(50) The second arm **4244** connects the first arm **4242** and the third arm **4246**. The second arm **4244** has one end coupled to the first arm **4242** and the other end coupled to the third arm **4246**. One end of the second arm **4244** is coupled to the other end of the first arm **4242**. The second arm **4244** may have a bent shape. The second arm **4244** has a first portion having a longitudinal direction corresponding to the longitudinal direction of the chuck pin **3446** and coupled to the first arm **4242**, a second portion extending from the first portion and extending in a direction perpendicular to the longitudinal direction of the chuck pin **3446**, and a bent portion connecting the first portion and the second portion. A hinge coupling portion hinge-coupled to the third arm **4246** may be provided on the second portion of the second arm **4244**.

(51) The third arm **4246** connects the second arm **4244** and the first magnetic body **422**. One end of the third arm **4246** is hinge-coupled to the second arm **4244**, and the other end of the third arm **4246** is hinge-coupled to the first magnetic body **422**. One end of the third arm **4246** is hinged to a hinge coupling portion provided on the second portion of the second arm **4244**.

(52) The second arm **4244** may be provided with an elastic member **426**. The elastic member **426** is elastically coupled to the second portion of the second arm **4244**. The elastic member **426** is provided between the end of the second arm **4244** and the inner wall of the spin chuck **342**. The elastic member **426** provides recovery force so that the chuck pin **346** moves from the open position to the contact position.

(53) The second driving module **440** is spaced apart from the first driving module **420**. The second driving module **440** does not come into contact with the first driving module **420**. The second driving module **440** faces the first driving module **420**. A partition wall is positioned between the second driving module **440** and the first driving module **420**. The partition wall may be a bottom wall of the spin chuck **342**. The partition wall may be a bottom wall of the extension part **3422** in which the first driving module **420** is accommodated. The downward movement of the first magnetic body **422** of the first driving module **420** is restricted by the partition wall. The upward movement of the second magnetic body **442** of the second driving module **440** is restricted by the partition wall. The second driving module **440** is provided outside the spin chuck **342**. The second driving module **440** does not rotate together with the spin chuck **342**.

(54) The second driving module **440** includes a second magnetic body **442** and a driver **444**. The second magnetic body **442** faces the first magnetic body **422**. The second magnetic body **442** faces the first magnetic body **422**. The second magnetic body **442** is spaced apart from the first magnetic body **422** by a predetermined distance. The second magnetic body **442** is provided as a permanent magnet. The second magnetic body **442** is provided to have a polarity opposite to that of the first

magnetic body **422**. For example, when the N polarity is disposed on the upper portion of the second magnetic body **442**, the N polarity is disposed on the lower portion of the first magnetic body **422**, and when the S polarity is disposed on the upper portion of the second magnetic body **442**, the S polarity is disposed on the lower portion of the first magnetic body **422**. Repulsive force acts between the second magnetic body **442** and the first magnetic body **422**. Referring to FIG. **4**, the second magnetic body **442** is provided in a ring shape.

(55) The driver **444** moves the second magnetic body **442**. The driver **444** moves the second magnetic body **442** in the direction of the rotation axis of the spin chuck **342**. The driver **444** moves the second magnetic body **442** in the vertical direction along the rotation axis direction of the spin chuck **342**. The driver **444** may be provided as any one of a cylinder, a stepping motor, a servo motor, and a solenoid coil. However, the present invention is not limited thereto, and any driving device capable of moving the second magnetic body **442** in the vertical direction may be applied.

(56) A plurality of first driving modules **420** may be provided. The first driving module **420** may be provided in the same number as that of the chuck pins **346**. One second driving module **440** may be provided.

(57) The first driving module **420** moves the position of the chuck pin **346** according to the change in the position of the second magnetic body **442**. Hereinafter, a process of moving the chuck pin **346** through the first and second driving modules **420** and **440** will be described.

(58) Referring to FIGS. **5** and **6**, the chuck pin **346** may be moved from the contact position in which the chuck pin **346** is in contact with the side portion of the substrate W to the open position in which the chuck pin **346** is spaced apart from the side portion of the substrate W. The driver **444** moves the second magnetic body **442** in an upward direction. When the second magnetic body **442** is moved upwardly, the spaced distance between the first magnetic body **422** and the second magnetic body **442** (hereinafter, a first gap d**1**) becomes narrower. When the first gap d**1** between the first magnetic body **422** and the second magnetic body **442** is narrowed to a predetermined gap (hereinafter, a second gap d**2**), repulsive force is generated between the first magnetic body **422** and the second magnetic body **442**. The first gap d**1** is a distance at which repulsive force is not generated between the first magnetic body **422** and the second magnetic body **442** which face each other, and the second gap d**2** means the distance in which repulsive force acts between the first magnetic body **422** and the second magnetic body **442**. The second gap d**2** may be greater than the first gap d**1**. When repulsive force acts between the first magnetic body **422** and the second magnetic body **442**, the first magnetic body **422** moves upwardly by the repulsive force. When the first magnetic body **422** is moved upwardly, the other end of the third arm **4246** hinge-coupled with the first magnetic body **422** is moved upwardly, and one end of the third arm **4246** coupled to the second arm **4244** is moved in a direction away from the central axis of the spin chuck **342** (outward direction). At this time, the elastic member **426** is compressed in a direction away from the central axis of the spin chuck **346**. As the third arm **4246** moves in the outward direction, the second arm **4244** also moves in the outward direction, and the first arm **4242** coupled to the second arm **4244** also moves in the outward direction to move the chuck pin **346** to the open position.

(59) Referring to FIGS. **5** and **7**, the chuck pin **346** may be moved from an open position spaced apart from the side portion of the substrate W to a contact position in which the chuck pin **346** is in contact with the side portion of the substrate W. The driver **444** moves the second magnetic body **442** in a downward direction. As the second magnetic body **442** moves downward, the first magnetic body **422** and the second magnetic body **442** are spaced apart from each other by a first distance d**1** or more. In this case, the repulsive force between the first magnetic body **422** and the second magnetic body **442** does not act. When the repulsive force between the first magnetic body **422** and the second magnetic body **442** is extinguished, the first magnetic body **422** is no longer moved upward, the second arm **4244** is moved in a direction (inward direction) toward the central axis of the spin chuck **342** by the restoring force of the elastic member **426**, and the first arm **4242** coupled to the second arm **4244** is also moved inward, so that the chuck pin **346** is moved to the

contact position to grip the side portion of the substrate W. In addition, when the second arm **4244** is moved inwardly by the restoring force of the elastic member **426**, one end of the third arm **4246** hinged with the second arm **4244** is moved inwardly, and the other end of the third arm **4246** hinged with the second magnetic body **442** is moved downwardly. The movement of the first magnetic body **422** in the downward direction is restricted by the partition wall positioned between the first magnetic body **422** and the second magnetic body **424**.

(60) Referring back to FIG. **2**, the substrate treating apparatus **300** includes a heating unit **500**. The heating unit **500** is a configuration for emitting a laser beam to the substrate W. The heating unit **500** may be positioned at a lower surface than the window member **348** in the substrate support unit **340**. The heating unit **500** may emit a laser beam toward the substrate W positioned on the substrate support unit **340**. The laser beam emitted from the heating unit **500** may pass through the window member **348** of the substrate support unit **340** to be emitted onto the substrate W. Accordingly, the substrate W may be heated to a set temperature. The heating unit **500** may include a laser beam generating member (not illustrated), a laser beam emitting member **500**, and a laser beam transmitting member. The laser beam generating member (not illustrated) may be disposed outside the chamber **310**. The laser beam emitting member **500** may include, for example, a lens. The laser beam emitting member **500** may include a plurality of lenses. The laser beam emitting member **500** may be provided inside the spin chuck **342**. The laser beam emitting member may be provided as a line connecting the laser beam generating member (not illustrated) and the laser beam emitting member **500**. The laser beam transmitting member may transmit the laser beam generated from the laser beam generating member (not illustrated) to the laser beam emitting member **500**. The laser beam emitting member **500** receiving the laser beam by the laser beam transmitting member may emit the laser beam to the substrate W. In the above, the heating unit **500** has been described as a configuration for emitting a laser beam, but the present invention is not limited thereto, and the heating unit **500** may be provided as a variety of heat sources capable of heating the substrate W. For example, the heating unit **500** may be provided as an LED or a halogen heater.

(61) Hereinafter, a substrate treating method according to an exemplary embodiment of the present invention will be described with reference to FIGS. **9** to **14**. FIG. **9** is a flowchart of a substrate treating method according to an exemplary embodiment of the present invention, and FIGS. **10** to **13** are diagrams sequentially illustrating the substrate treating method of FIG. **9**.

(62) FIG. **10** is a diagram illustrating an alignment operation of a substrate before starting a process according to the substrate treating method according to the exemplary embodiment of the present invention. Referring to FIG. **10**, before the process starts, the substrate W is transferred to the substrate support unit **340**. After the substrate W is transferred to the upper portion of the spin chuck **342**, the center of the substrate W is aligned with the center of the spin chuck **342** or the window member **348**. In this case, the chuck pin **346** is positioned at the open position. When the center of the substrate W is aligned with the center of the spin chuck **342** or the window member **348**, the chuck pin **346** is moved to the contact position to support the side portion of the substrate W. In this case, the chuck pin **346** is moved by the chuck pin moving unit **400**.

(63) FIG. **11** illustrates a process of forming a processing liquid puddle on the substrate according to the substrate treating method according to the exemplary embodiment of the present invention. Referring to FIG. **11**, the spin chuck **342** rotates the substrate W at a first speed in the state where the substrate W is supported by the chuck pins **346** of the substrate support unit **340**. The liquid supply unit **390** forms a first liquid film C**1** on the upper surface of the substrate W by supplying a first liquid to the substrate W rotating at the first speed. The first liquid film C**1** may be a puddle having a predetermined thickness. When the first liquid is supplied to the substrate W, the chuck pin **346** is positioned at the open position. That is, the first liquid film C**1** is formed by supplying the first liquid to the substrate W rotating at the first speed in the stat e where the chuck pin **346** is positioned at the open position in which the chuck pin **346** is spaced apart from the side portion of the substrate W. When the chuck pin **346** is positioned at a contact position in contact with the side

of the substrate W when the first liquid film is formed, there is a problem in that the first liquid flows down along the chuck pin **346** and it is difficult to maintain a certain amount of the liquid film. However, according to the substrate treating method according to the exemplary embodiment of the present invention, when the first liquid film C**1** is formed, the first liquid film C**1** is formed in the state in which the chuck pin **346** is positioned at the open position in which the chuck pin **346** is spaced apart from the side portion of the substrate W, so that it is possible to prevent the flow-down phenomenon of the first liquid, thereby maintaining a certain amount of the liquid film. In addition, as the chuck pin **346** is spaced apart from the side portion of the substrate W, there is an advantage in that the first liquid film (puddle) may be easily formed by the surface tension of the first liquid film C**1**.

(64) FIG. **12** is a diagram illustrating an operation of heating the first liquid film according to the substrate treating method according to the exemplary embodiment of the present invention. Referring to FIG. **12**, when the first liquid film having a predetermined thickness is formed, the rotation of the substrate W is stopped and the supply of the first liquid from the liquid supply unit **390** is stopped. As a result, the surface of the substrate W is covered with the first liquid film C (formation of the puddle of the processing liquid). Here, the processing liquid may be provided as an aqueous solution of phosphoric acid. The substrate W and the first liquid film C**1** are heated to a temperature suitable for processing the substrate W. The substrate W and the first liquid film C**1** may be heated by the heating unit **500**. At this time, the chuck pin **346** is located at the open position. As a heat source of the heating unit **500**, any one of a laser, an LED, and a halogen heater may be provided. By maintaining the state in which the surface of the substrate W is covered with the first liquid film C**1** of the heated processing liquid for a predetermined period of time, the treatment of the substrate W (for example, etching processing (wet etching process)) is performed. At this time, the spin chuck **342** rotates the substrate W at the first speed. The first liquid on the substrate W may be stirred by rotating the substrate W about half rotation in the forward rotation and reverse rotation direction. Thereby, etching is facilitated, and the in-plane uniformity of etching amount may be improved.

(65) FIG. **13** is a diagram illustrating a second liquid supply operation according to the exemplary embodiment of the present invention. Referring to FIG. **13**, after the first liquid film C**1** is heated, a second liquid is supplied to the substrate W to form a second liquid film C**2**. In this case, the spin chuck **346** rotates the substrate W at a second speed. In addition, the chuck pin moving unit **400** moves the chuck pin **346** to the contact position in which the chuck pin **346** is in contact with the side portion of the substrate W. The second liquid may be provided in the same liquid as the first liquid. The second liquid may be provided as an aqueous solution of phosphoric acid. The thickness of the second liquid film C**2** may be smaller than the thickness of the first liquid film C**1**. The second speed may be faster than the first speed. For example, the rotation at the first speed may be a low-speed rotation, and the rotation at the second speed may be a high-speed rotation. For example, the first speed may be equal to or less than 20 RPM.

(66) The substrate treatment according to FIGS. **10** to **13** may be repeatedly performed a plurality of times. When the substrate treatment according to FIGS. **10** to **13** is finished, rinse processing (rinsing process) of supplying a rinse liquid to the substrate W to remove by-products generated by reaction with the processing liquid from the surface of the substrate W is performed. The rinse liquid may be pure water (DIW).

(67) According to the substrate treating method according to the exemplary embodiment of the present invention, in the first liquid supply operation and the liquid film heating operation, the chuck pin **346** is moved so that the chuck pin **346** is positioned at the open position, and in the second liquid supply operation, the chuck pin **346** is moved so that the chuck pin **346** is positioned at the contact position. At this time, according to the general spin chuck structure of moving the chuck pin between the open position and the contact position by using a rotary cylinder that can rotate 90 degrees, the chuck pin is movable only when the spin chuck is stopped. Therefore, when

the spin chuck has to be stopped and the chuck pin has to be moved whenever each operation of the substrate treating method is performed, the process takes a long time and the process efficiency is degraded, and when the spin chuck is stopped for the stop of the spin chuck and the movement of the chuck pin at each process operation, a liquid film having a uniform thickness may not be obtained, or the liquid film may be excessively hardened, causing the liquid film to break. In addition, when the treatment the substrate progresses in a state in which the chuck pin is in contact with the side portion of the substrate without stopping the spin chuck at each operation of the substrate treating method in the spin chuck structure in the related art, as described above, there are problems in that the liquid film flows along the surface of the chuck pin and thus a certain amount of liquid film cannot be formed, and since the surface tension of the liquid film is reduced by the contact area between the chuck pin and the side portion of the substrate, it is difficult to form a liquid film of a certain thickness.
(68) However, according to the exemplary embodiment of the present invention, it is possible to move the chuck pin **346** even during the rotation of the spin chuck **342** through a chucking system utilizing the repulsive force of the rotating first magnetic body **422** and the non-rotating second magnetic body **442**, which are spaced apart from each other, thereby solving the above-mentioned problems.
(69) Meanwhile, the substrate treating apparatus and the substrate treating method according to the above-described exemplary embodiments may be controlled and performed by a controller (not illustrated). Configuration, storage and management of the controller may be realized in the form of hardware, software, or a combination of hardware and software. The file data and/or the software configuring the controller may be stored in volatile or non-volatile storage devices, such as Read Only Memory (ROM); or memory, such as, for example, Random Access Memory (RAM), memory chips, devices, or integrated circuits, or a storage medium, such as Compact Disk (CD), Digital Versatile Disc (DVD), magnetic disk, or magnetic tape, which are optically or magnetically recordable and simultaneously machine (for example, computer)-readable.
(70) The foregoing detailed description illustrates the present invention. In addition, the foregoing is intended to describe exemplary or various exemplary embodiments for implementing the technical spirit of the present invention, and the present invention may be used in various other combinations, changes, and environments. That is, the foregoing content may be modified or corrected within the scope of the concept of the invention disclosed in the present specification, the scope equivalent to that of the disclosure, and/or the scope of the skill or knowledge in the art. Accordingly, the detailed description of the invention above is not intended to limit the invention to the disclosed exemplary embodiment. In addition, the appended claims should be construed to include other exemplary embodiments as well. Such modified exemplary embodiments should not be separately understood from the technical spirit or prospects of the present invention.

Claims

1. An apparatus for treating a substrate, the apparatus comprising: a processing vessel having a processing space; a support unit configured to support the substrate in the processing space and rotate the substrate; a liquid supply unit configured to supply a processing liquid to the substrate supported by the support unit; and a heating unit configured to heat the substrate, wherein the support unit includes: a spin chuck; a driver configured to rotate the spin chuck; a chuck pin installed on the spin chuck so as to be rotated together with the spin chuck; a window member configured to be disposed under the substrate, wherein the window member includes a hole within which the chuck pin is disposed and wherein the window member is configured to be shaped to match a shape of the substrate; and a chuck pin moving unit configured to move the chuck pin between a contact position in which the chuck pin is in contact with a side portion of the substrate and an open position in which the chuck pin is spaced apart from the side portion of the substrate,

wherein the chuck pin moving unit includes: a first driving module coupled to the chuck pin and rotated together with the spin chuck; and a second driving module which faces the first driving module and is not rotated together with the spin chuck, wherein the first driving module includes: a first magnetic body; and an arm member connecting the first magnetic body and the chuck pin, wherein the arm member is configured to guide the movement of the chuck pin as the first magnetic body moves, and wherein the arm member includes: a first arm coupled to the chuck pin and extending in a direction perpendicular to a longitudinal direction of the chuck pin; a second arm coupled to the first arm; and a third arm configured to connect the second arm and the first magnetic body, wherein the third arm is hinge-coupled to each of the second arm and the first magnetic body, wherein an elastic member is coupled to one end of the second arm, and wherein the elastic member is configured to provide recovery force so that the chuck pin is moved from the open position to the contact position, wherein the second driving module includes a second magnetic body facing the first magnetic body, and a driving member configured to drive the second magnetic body in a vertical direction, and wherein a space where the second drive module is disposed and a space where the first magnetic body is disposed are separated from each other by a bottom of the spin chuck.

2. The apparatus of claim 1, wherein the first driving module moves a position of the chuck pin according to a change in a position of the second magnetic body.

3. The apparatus of claim 1, wherein a repulsive force acts between the first magnetic body and the second magnetic body.

4. The apparatus of claim 1, wherein the chuck pin includes a plurality of chuck pins, the second magnetic body is provided in a ring shape, and the first driving module is provided in a number corresponding to the plurality of chuck pins.

5. The apparatus of claim 1, wherein the chuck pin moving unit moves the chuck pin while the spin chuck is rotating.

6. The apparatus of claim 5, wherein the driver rotates the spin chuck so that the substrate is rotated at a first speed, and the chuck pin moving unit moves the chuck pin so that the chuck pin is positioned at the open position while the substrate is rotating at the first speed.

7. The apparatus of claim 6, wherein the driver rotates the spin chuck so that the substrate is rotated at a second speed faster than the first speed, and the chuck pin moving unit moves the chuck pin so that the chuck pin is positioned at the contact position while the substrate is rotating at the second speed.

8. The apparatus of claim 1, wherein the spin chuck has a through-hole penetrating in a vertical direction, and the heating unit heats a bottom surface of the substrate through the through-hole.

9. The apparatus of claim 8, wherein the heating unit includes a laser.

10. The apparatus of claim 8, wherein the spin chuck includes: a body portion; and an extension portion extending upwardly from an upper end of the body portion, wherein an area of the extension portion gradually increases toward the top.